A Field Guide to Mathematics at Southampton

1st Year Student (25/26)

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WHAT THIS GUIDE IS (AND ISN'T)

> Mathematics is one of the most beautiful things in the world and it brings joy and fulfilment and frustration in equal measure, and what more could you possibly want out of life
>
> *Graham A. Niblo*

This is not a Maths textbook.

It won't teach you how to compute an integral or prove a theorem step by step. It won't replace lectures, problem sheets, or workshops, and it isn't a shortcut through the degree.

Think of it instead as a transition guide, written from inside the experience and not from the other side of it.

There are countless books and resources about transitioning to university mathematics. However, most I found were written by people who had already finished the journey and looked back with clarity and hindsight, so I wanted something written from inside the process, something honest that reflects what progress feels like while it's happening.

This guide is written by a student who is still in the process of learning how to do university mathematics. That matters. It means the reflections here are not polished by years of hindsight, but shaped by recent confusion, breakthroughs,

stress before exams, and moments where something finally clicked.

There is a difference between having mathematical tools and knowing how to use them in an unfamiliar territory. At school, I often felt like I was collecting tools, and when I came to university, I realised I didn't always know how to use them.

This is not comprehensive, it does not cover every module or every scenario. It is simply an attempt to articulate the parts of the degree that are rarely written down: the mindset shift, the moments of doubt, the practical habits that make things easier, and the difference between understanding something and merely recognising it.

In the chapters that follow, I'll talk about:

- The shift from school mathematics to university mathematics
- What lectures, coursework, and workshops are really for
- How to approach problem solving when you don't immediately know what to do
- The temptation to outsource thinking, and what that costs
- Revision, exams, and time management
- Asking for help, giving useful feedback, and making the most of helpful resources

This guide isn't here to solve all possible problems you may encounter, but it's here to help you think about how you learn. You don't need to read it all at once. Some chapters might resonate immediately, others might make more sense later.

University mathematics is demanding, but also deeply rewarding. And this guide is simply an attempt to make the path into it a little clearer.

I've also scattered a few **xkcd** references along the way because sometimes, a well-timed joke explains the mood of mathematics better than a paragraph every could.

Not everyone signs up for a maths degree wanting a philosophical reflection on learning, and that's completely fine. But if you want to care about how you learn, and not just what you learn, I hope this helps.

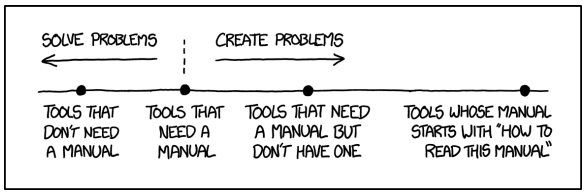

Figure 1: **Manuels**

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THE JUMP

> I was born not knowing and have only had a little time to change that
>
> *Richard P. Feynman*

1.1 The Mindset Shift

At school, I was used to understanding things quickly.

If I didn't understand something in class, I usually would by the end of the lesson. And if I didn't, I could go home, do a few questions, and it would click. There was a clear structure, you learned a method, applied it, and got better.

University maths felt different almost immediately.

Having worked at a department open days, one of the most common questions I get asked is 'What's the jump actually like? Is it just harder? Is it just more content?'

My honest answer has been and still is: it's just different.

One of the most common issues that students face, and this is more than just my personal experience, is that they no longer understand things straight away. This was something which I recall being addressed on numerous university open days when I was still in school, but I didn't truly understood what that meant until I got to university.

Often at school, not understanding usually meant you needed more practice. However at university, not understanding often meant to me that I needed more time, or more reflection, or just a shift in how I was used to thinking. This is especially true if you are used to being the 'the maths person', and used to being the one who got it first. The point I want to convey is that the jump isn't just academic, it's also psychological.

1.2 Why Maths is Different

In many subjects or courses, you can revise by memorising key ideas, reading around the topic or absorbing information repeatedly until it sticks.

Mathematics doesn't work like that. You can't (always) memorise your way into deep understanding or skim a proof and expect it to feel natural. You can't treat definitions and just some vocabulary words and hope they'll make sense.

One of the best things I like about university maths is that it shifts the focus from 'doing' to 'justifying'. You go from merely applying formulas to understanding why those formulas exist in the firs place, and from being shown the path to a theorem to learning how to build the path yourself.

You might be used to mastering a particular technique or a method from school, and those skills continue to be useful when you transition to university, but more often than not, you find yourself exploring structure. And exploration is slower, but also more rewarding.

1.3 Redefining Understanding

The underlying issue in my experience is often to do with expectation students have from themselves rather than their level of intelligence.

Mathematics is a conceptually difficult subject to grasp. If you approach each lecture or assignment with the mindset that you're expected to understand everything, or feeling confused means you're not good enough, the delay or lack of those things can feel like failure.

Having such expectations from yourself are completely understandable and they might've worked well before, but you might find that they don't work in the same way now. Maths is build on layers. You might understand the notation before the idea, or follow a proof before you really see why it's necessary.

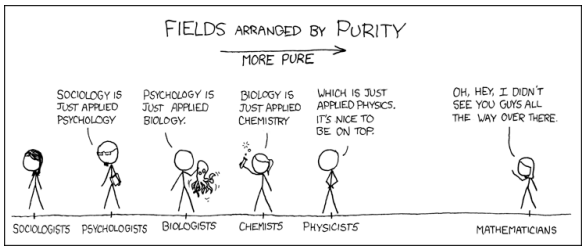

Figure 1.2: **Purity**

If you ever feel confused or find yourself struggling, rest assured, you're not alone. It's incredibly common, normal, and all part of the process.

Now for the challenging part, you have to let go of the need to understand everything immediately. This doesn't mean lowering standards, but adjusting expectations as discussed in the section above.

Understanding at the level at which university maths progresses takes time. It requires sitting with new ideas and concepts you might've never seen before. Trying things that don't work, and being okay with not being the fastest or the brightest in the room. It's less about speed and more about structure.

The jump isn't just from 'easy' to 'hard'. I think it's more from procedural to conceptional, from the methods you might be familiar with, to the mindset shift you might have to make. And once you accept that, the confusion will hopefully start to feel less threatening.

LECTURES AND COURSEWORK

> The brain is a wonderful organ; it starts working the moment you get up in the morning and does not stop until you get into the office
>
> *Robert Frost*

2.1 Overview

Each year, you will study four modules each semester (excluding Core Skills), so eight across both semesters. Each module has three lectures per week lasting about 45 minutes, along with problem classes.

Lecturers will normally provide lecture notes or slides beforehand, which can be found on Blackboard. It's worth having a quick look at these before the lecture if you can, even just seeing the headings can make it much easier to follow what's happening when you're sitting in the lecture hall.

The important thing to remember is that lectures are only one part of the learning process. They introduce the ideas, but most of the real understanding comes later when you spend time thinking about the material yourself, working through problems, and discussing things with other students.

You'll become familiar with two names very quickly: **Blackboard** and **Seats**.

Blackboard is the university's online learning platform where lectures notes are uploaded, announcements are posted and the coursework and recordings are stored. If something official happens in the module, it will most likely appear there.

One of the most useful things you can do in the first week is link your timetable to your own calender, whether that's Google calender or something equivalent. Remember that universities will not chase you the way school did. There are no bells and no-one reminds you every morning where you're supposed to me, so the responsibility shifts to you.

Seats is the system used for timetabling and attendance. It shows your lecture locations, times, and room numbers. It's also how attendance is recorded. At the start of lectures, a QR or Seats code will be displayed on the board which students can scan or type to register their attendance on the app.

It is worth getting into the habit of checking both early in the semester because you might find it may unnecessary stress. Knowing where things are in advance means you can focus on the maths rather than the logistics.

2.1.1 Core Skills

2.1.2 Problem Classes

Each module will have a weekly problem class. These are quite different from lectures. Instead of listening to new material being introduced, the focus is on working through questions and understanding how the ideas from the lectures actually get used.

The exact format can vary slightly between modules and lecturers. Sometimes the lecturer will go through selected problems on the board, sometimes students will suggest questions they want to see explained, and sometimes you go might go through some coursework questions.

One of the most important thing you can do to make problem classes useful is look at the problems beforehand. If you walk into a problem class having never seen the sheet before, it's much harder to follow what's going on and becomes easy to just copy down solutions without really understanding them.

2.2 Recorded Doesn't Mean Optional

You will quickly realise that most lectures are recorded, and at some point you might be temped to think, 'I can just watch it later', especially on a cold and rainy

morning, or after a late night, or when the idea of sitting in a lecture feels mildly exhausting.

I cannot stress this enough, but being physically present in a lecture matters more than you think. When you are in the room, you are less likely to pause a video and get distracted by your phone. You are also less likely to tell yourself that you'll 'finish it later' and risk not coming back to it. You are more likely to stay with the explanations as they unfold, even if you don't fully understand them yet. You hear the throwaway comments that never make it into typed notes, and pick up on the emphasis in the lecturer's voice when they say something is important, and that emphasis stays with you.

There is also something about the collective focus that recordings cannot replicate. Sitting in a room full of people who are all attempting to follow the same proof creates a concentration that watching it alone at 1.5x speed, half distracted cannot compete with. In addition to that, you are able to ask questions during or after the lecture if you attend in person.

Recordings are useful, in fact, they are incredibly useful for revision or revisiting a proof if that felt unclear the first time. They are a safety net, but not a replacement for showing up.

That said, there will be weeks where life happens. Missing a lecture isn't the end of the world. Just don't let 'I'll watch it later' turn into 'I'm now three weeks behind and slightly afraid to look at the video'. The longer you rely on your future self to catch up, the easier it becomes to drift out of sync with the module, and once you're out of sync, everything feels harder than it needs to be.

So if you can be there, be there.

2.3 When Your Brain Just Isn't Working

There often comes a point in almost every student's journey where one morning, for no dramatic reason, your brain simply refuses to cooperate.

The lecturer is speaking, and the words are technically English, and the notation seems familiar. The content might not even be complicated, and yet, nothing seems to land. You realise you've copied half a page without actually processing a single line and catch yourself been thinking about what you're going to have for lunch for the last 8 minutes.

Rest assured, this is normal and happens to the best of us. It does not mean you

have suddenly lost the ability to think mathematically. Engaging with university mathematics requires sustained concentration in a way that school rarely did. The abstraction is higher and the cognitive load is heavier. Your brain will sometimes hit its limit, especially if you're tired, stressed, slightly underslept, or just adjusting to a new routine and environment.

Not every lecture will feel inspiring and not every topic will immediately excite you. On days when your performance is nowhere near optimal, lower the bar slightly. Instead of aiming to understand everything, narrow the target. Pick one line or sentence on the board and ask yourself what it is actually saying. Or write down one question you would ask if you were brave enough, or underline one step in the proof that feels unclear and mark it to revisit later.

If your attention in slipping, change your posture. Sit up straight and just listen without feeling the need to copy everything down. Just write a short sentence that summarises what you have learned so far, or questions if you're confused about something. Small physical adjustments often help reset focus more than you'd expect.

And if you're brain truly isn't cooperating despite your best efforts, just accept that for what it is. Stay present as best you can, then review the material later when you're more alert. Not all the understanding needs to happen in one sitting in the lecture hall.

The key message here being, one unfocused lecture does not define your trajectory[1]. Boredom, distraction, fatigue- it's all part of the learning experience, and learning to work with them rather than against them is part of becoming a good learner.

2.3.1 Catching Up On Lectures

At some point, you may very well miss a lecture. Maybe you were ill, overslept, or just needed a morning off. It happens, and one missed lecture is not a crisis.

What turns it into a crisis is delay.

One missed lecture feels manageable, two feel slightly annoying and by week five, you might start avoiding the recordings or looking at notes entirely because it just feels overwhelming.

The trick to avoid going down that spiral is quite simple: catch up quickly. If you miss a lecture, watch the recording within a day or two while the topic is still current and notation still looks familiar. Don't let it all pile onto future you. If for some

[1] Pun intended

reason due to time constraints you can't watch the recordings, ask your coursemates or friends about which point in the notes the lecturer has gone to and read up to that point.

When you do catch up, don't treat the recording like a background noise. Pause and try to follow the argument yourself. When a definition appears, stop and ask whether you understand what each part is doing and when a proof begins, try and predict the next step before the lecturer reveals it.

Falling slightly behind is very normal. But letting it snowball is what makes learning more difficult than it needs to be.

2.4 Asking questions

There is often a quiet fear around asking questions in lectures. If you ever find yourself sitting in a lecture room, slightly unsure about a definition or confused about some step in a proof, and you look around the room and everyone else looks calm, and you might think, 'Maybe it's just me'.

It almost never is. Most people are running through the same internal debate of 'Is this worth asking? Is this obvious? Is this too stupid of a question?', etc.

Here's something I've learned: a question does not need to be clever, it just needs to be honest. Questions like 'How did we go from Line 1 to Line 2 of this proof', or 'Is there an example where this conditions fails?' are perfectly good questions to ask in lecture. There are no stupid questions.

By asking questions, you not only benefit yourself, but also others in the lecture, including the lecturer themselves. The lecturer can see where the gaps are, other students relax because someone has said what they were thinking, and most importantly, the room becomes less passive. Learning mathematics, doing mathematics, is a process, and asking questions is part of that process.

If asking during the lecture feels too intimidating, most lecturer's are more than happy for students to ask them at the end of the lecture, or during office hours and workshops.

2.4.1 A note on talking in lectures

Lectures are shared learning spaces and everyone in the room is trying to concentrate on the same material.

It might seem harmless to chat quietly during a lecture, but even small conversations can be a distracting for the people sitting nearby especially with an attention demanding subject like maths.

More importantly, it's also a matter of respect. The lecturer has prepared the material and everyone else in the room has chosen to be there to learn. Talking over them or holding side conversations can come across as rude even if that isn't the intention.

If you want to discuss something with the person next to you, it's usually better to wait for a natural pause, write it down somewhere, or bring it up after the lecture. When everyone treats the lecture space with a bit of consideration, it makes it much easier for everyone else to follow what's going on.

2.5 Giving feedback

At some point during the semester, you will be asked to fill in feedback forms for SSLC or mid-semester.

It is very tempting to either rush through them, or completely ignore them. It is also very easy to write things like 'Too fast' or 'Confusing' or 'Good'.

But feedback only becomes useful when it is specific. Writing 'This lecture is confusing' doesn't give anyone anything to work with, whereas writing 'There are too many definitions and very few examples to see how those definitions work' is useful.

Some responses convey reaction, others information, so ensure that whenever you're in a position of providing feedback, you relay the information over reaction as best as you can.

Lecturers are not mind readers. They often assume a certain level of clarity because they've been living with the material for far far longer than we have. So if ten students feel quietly lost but say nothing, nothing changes.

Feedback not only helps the lecturers, but also the cohort. And it helps the individual, because the act of explaining why something was unclear forces you to reflect on what you were expecting.

2.5.1 Too fast? Too Slow?

There will be lectures where some definitions will arrive in rapid succession and you are still trying to process the first line while the theorem is already on the board.

A lecture being 'fast paced' usually happens when you are encountering abstraction for the first time, which takes time to settle. Exposure of a new topic precedes clarity, and you are not expected to internalise everything immediately.

On the other hand, there might be lectures that feel slow, perhaps even repetitive, and you may find yourself zoning out because you think you already understand it. That is often the moment where it is worth looking deeper into what it being taught, asking why the definition is phrased in a certain way, or what breaks if an assumption is removed, or even how the idea connects to something you've already seen.

The pace of a lecture is rarely perfectly aligned with your personal understanding, and that is normal. However, if you feel like something is unreasonable, you should mention that in either one of the feedback forms or by talking to your lecturer.

2.6 Coursework

Most modules will have some form of coursework, which in practice usually means a weekly problem sheet or set of exercises that are released on a fixed day and due the following week. For example, you might get an assignment set on Monday afternoon and due the next Monday, but always check the specific module outline for the details. Some coursework might be on paper, others might be delivered using an online environment like Stack.

At first, coursework can feel constant. You submit one sheet and the next one appears almost immediately, and it can feel like there is never a complete clear week. Having said that, the regular rhythm of doing problem sheets is actually really helpful as it stops the module from going into the 'I'll catch you later' territory and encourages you to engage with the content.

The key is to treat coursework as part of your weekly structure instead of something extra that appears unexpectedly. Build your week and routine around when the assignments are due, and decide early when you will attempt each one.

More often than not, the hardest part of coursework is getting started with it. Start working on it earlier than you think. There can be a very common pattern in the first year where the coursework is released on Monday, you glance at it on Wednesday, and realise on Sunday evening that you still haven't attempted exercise 1.

If you leave assignments until the night before, you reduce the amount of time you have to get stuck productively. The questions are designed to make you think, and thinking takes time.

So try and get into a routine of starting the coursework at least one day in advance if not before. If you have attempted a question and still feel stuck, the department has built something very useful into the structure of the week that helps with such problems: **First Year Workshops**. For if and when find yourself on coursework, the next chapter aims to explain how workshops work.

2.6.1 Extensions and Special Considerations

As with everything, you might have moments where multiple deadlines collide in a way that genuinely overwhelms you. Universities know this and that's why extensions and special consideration procedures exist.

WORKSHOPS AND OTHER RESOURCES

> To doubt everything or to believe everything are two equally convenient solutions; both dispense with the necessity of reflection.
>
> *Henri Poincaré*

3.1 First Year Mathematics Workshops

3.1.1 Introduction to Workshops

3.1.2 More Than Just Coursework Help

When I first started, I thought the first year workshops were basically just getting help for your coursework, and nothing else.

But that would be missing a major point. Workshops are not there to replace your thinking, but to refine it.

The biggest difference I noticed was that when I genuinely struggled with a problem, even if I got nowhere, the workshop became the most valuable resource. If you go hoping someone will just show you the solution from scratch, you leave having learned much less than you could have. Even 15 minutes of serious attempt make

major impact. It makes you aware of where you're stuck, and gives you something specific to ask about.

3.2 AI and Outsourcing Thinking

It would be highly ignorant and unrealistic of me to not talk about Generative AI. We all know it exists and it can generate solutions instantly. And sometimes, it is useful, it can clarify notation or rephrase definitions and help fill in some gaps in reasoning.

However, the biggest drawback of AI is that it removes the part of the process that actually builds the understanding. If you ask AI to generate a full solution to a problem you haven't seriously attempted, you might feel productive, even pleased looking at the solution, but you've just skipped the uncomfortable but crucial bit where your brain learns by trying, failing, and reattempting ideas. And that's the uncomfortable bit where understanding takes place.

There is a difference between using AI to complement your learning, and using it to replace your work and your thinking entirely, and it is important that you be honest with yourself about which one you're doing. At the end of the day, you are the one being impacted by your use/misuse of AI, so take accountability.

The department and the university has its own policy on AI, so I urge you to read it because it matters not only due to academic integrity, but because this degree is about learning to think independently.

PROBLEM SOLVING

> If you can't solve a problem, then there is an easier problem you can solve: find it.
>
> *George Pólya*

4.1 Building the Toolkit

In this chapter, I would like to challenge the common misconception that a Maths degree is nothing more than learning more advanced techniques.

If there's one thing I'd like you to remember while studying mathematics, it's this: the things you learn didn't appear out of nowhere.

Every definition, every theorem, every technique you come across exists because at some point someone was trying to solve a problem.

When you first encounter a new topic, it can sometimes feel like a wall of definitions followed by a list of results you're expected to accept and apply. And it's easy to walk away thinking that mathematics is just a collection of rules someone decided on a long time ago.

But that's not really how it developed.

Mathematicians in the past were trying to understand patterns, structures, and problems that genuinely puzzled them. The definitions and theorems you see

today are often the end result of a long process of people asking questions, trying ideas, getting stuck, and refining their thinking.

In other words, mathematics didn't start with the definitions. It started with curiosity.

Once you start viewing mathematical ideas as tools that were developed to tackle specific problems, things begin to feel a little less arbitrary. A theorem stops being just a statement you're supposed to memorise, and starts looking more like a clever idea someone came up with because it helped them get somewhere.

Of course, when you first learn a topic, you're usually seeing the polished version, the final structure that survived years of refinement. The messy exploration that produced it often isn't visible in a lecture course.

That can make the subject feel colder than it really is. But behind every theorem there was once a mathematician trying to make sense of something that didn't quite add up.

You might not immediately appreciate every idea in the same way they did, and that's completely normal. Some motivations only become clearer much later, when you see how different areas connect.

Still, it's worth remembering that the mathematics you're learning exists because someone, somewhere, cared enough about a problem to keep thinking about it.

And in a way, that's what you're joining when you study it. What this degree really trains is how you respond when you don't immediately know what to you.

At school, most problems you might've dealt with followed a recognisable pattern. You learned a method, applied it and improved with practice. At university, you will encounter questions that don't look like anything you've seen before. And that's deliberate.

Mathematics is not training you to memorise procedures, but training you to sit with uncertainty without panicking, to break a large problem into smaller parts and identify structures where none seems obvious in the first glance.

When faced with something unfamiliar, mathematics teaches you to ask:

- What do I acutally know?
- What is the definition here?
- What would have to be true for this to work?
- Can I test a simpler example with numbers before generalising it?

Those questions become automatic over time.

4.1.1 Notes That Actually Help

There seems to be a common myth that good notes always equal good understanding. Spoiler: they don't.

Note taking is not about transcription, but about processing. You might even spend the first few weeks trying to find the 'right' way to take notes for something like Mathematics.

Some people swear by handwriting because it slows them down, others type in LaTeX or something else because it might be neater and easier to organise, some annotate the lecture slides or notes directly.

There is no 'one-size-fits-all' here. What works brilliantly for one person might be useless for another. The only reliable test is whether the system helps you think critically or not.

If you find yourself with beautiful notes on the most recent Calculus lecture you attended but have no idea on how to approach the problem sheet, something needs to change. If you find not making any notes in lectures and reviewing information later in the evening helps ideas settle, try that. Or if you realise handwriting slows you down, see if you can find a way to condense information instead of copying a lot of things down.

University requires experimentation, not perfection. The goal here is not to produce the most elegant looking notes amongst your peers, but to build understanding that you can use for yourself. You are not being graded on your notes, but on your thinking, so ensure that the learning resources you use/build contribute to your thinking.

4.1.2 Explaining Things Out Loud

There is one habit that has helped me more than any particular note-taking method, and it has nothing to do with whether I typed or wrote by hand. It was explaining things out loud.

A good test to see if you've understood something is usually later in the day or in the week, close your notes and try to explain the main idea either to a friend or out loud.

You very quickly discover what you don't understand when you try to speak it.

On paper, everything can look fine. But the moment you try to explain why some assumption is necessary, or why an implication follows, you feel the gap immediately. And that gap is useful information, it tells you exactly where to focus.

If you can explain a concept to someone else, even to an imaginary audience without constantly checking your notes, you are much closer to understanding a concept than if you simply reproduce what was written in the lecture.

Mathematics is not about producing the most beautiful notes, it's about producing clarity and understanding of the concepts learned.

4.2 What to Do When You're Stuck

Problem solving sounds very inspiring until you're staring at Question 4 from Coursework 2 at 10:37pm or 38 minutes into the exam and your brain has gone offline. What I mean to say is, there will be moments in lectures or problem sheets or exams where you may genuinely not know how to begin.

The instinct when stuck is often to rush, to scan the question faster or hope something jumps out.

Instead, try and slow down. Read the question again. What is actually being asked? Is it existence? Bijectivity? Optimsation?

Rewrite the question in simpler terms and alongside it, any definitions or theorem you can recall about what is being asked.

This does two things. First, it forces your brain back into structure and something you do know rather than panic. And secondly, everything that you need is given to you in the words of the question, and once you have unpicked the most important parts, you have part of a path forward.

Often when you're trying to prove something, or show something holds, you want to look at the statement and ask yourself, 'what would have to be true for this to work¿ and work backwards from there. Working backwards from what you know is true can help you get started.

Over time, you might start noticing that the moments of being stuck becomes less frightening. Not because you always know what to do, but because you trust that you have strategies. And that trust is part of the toolkit.

4.3 On Gaining Insight

> One of the hardest things to unlearn at university is the urge to look at a solution the moment you feel stuck. Once you've seen someone else's solution, you can never recreate the benefit of discovering it yourself. Even if you don't solve the problem completely, struggling with it, forming wrong ideas, testing different approaches and hitting multiple dead ends does far more for your mathematical growth than neatly following a polished answer ever will.

To illustrate this with an example, there was a problem in my Linear Algebra II module where I was stuck on what, in hindsight, is a very simple result: show that the set V^S of all maps form a set S to a vector space V was a vector space. This sentence probably means very little unless you have already studied Linear Algebra, and that's fine. The technical details don't matter here.

What matters is what the problem requires. To show something is a vector space, we just need to check the definition we are given with the axioms and see if they are consistent. For this question, we only had to check that there exists an additive identity, every element has an additive inverse, and that one of the distributivity law holds.

When I first read the question, I understood the definition of an additive identity in theory. I knew what a 0_V meant in vector space, but I didn't know how to write the proof properly in this new setting, I was struggling with turning the idea in my head into a clean mathematically sound argument on paper.

Now, I could've easily looked up the proof or asked AI to generate it for me in a few seconds, and it would have probably told me to let $0(s) = 0_V, \ \forall s \in S$ then check $(f + 0)(s) = f(s)$. It would have been correct, looked neat, and I could have convinced myself I understood it.

But I didn't. Instead, I was in the workshop where I had my lecturer from my Linear Algebra II module present to talk me through the problem and refine the gaps in my thinking.

Instead of being simply told what I had to write for the proof to fall out, we started from what I did know about the definition of additive identity and what it was supposed to satisfy, namely the condition $f + 0 = f$, then treated this like an equation and worked out what this mysterious 'zero function' would look like.

So that day I learned that the zero element in that space of functions is just the function that sends every input to the zero vector. But arriving at that idea, and

more importantly, learning how to justify it properly, was the real insight.

What clicked in that workshop moment wasn't just the answer to one problem but how proofs in algebra work. You start from definitions, think about what must be true, and sometimes, work backwards to get a full picture of the proof before starting to write it.

The point being, that insight didn't come from seeing the finished proof. It came from wrestling with the gap between knowing the definition and knowing how to write it in the abstract way the questions asked me to. Now, if I had skipped that part and just read up the solution, I would have 'known' the answer, but would have lost the experience of learning how to reconstruct that kind of proof myself.

And that's the difference.

When you outsource your thinking, you might take pride in being incredibly efficient. But the price of efficiency is ownership, and the price of speed is insight, and with a subject as elegant and satisfying like Mathematics, that trade is just not worth it.

4.4 Why This Matters

This reason these problem solving skills matter is not because you will be constantly proving theorems outside university.

It matters because the habits transfer.

Learning how to decompose complex problems into manageable parts, staying calm under uncertainty, testing assumptions, communicating reasoning clearly. These are all thinking skills. A maths degree does not just teach content, but reshapes how you approach problems in general.

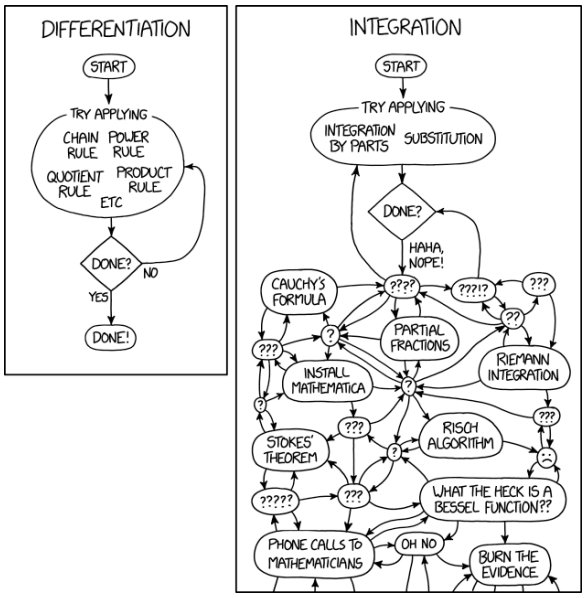

Figure 4.3: **Differentiation and Integration**

ALL ADVANCED MATH TECHNIQUES

Figure 4.4: **Advanced Techniques**

EXAMS

The epigraph for this chapter is left as an exercise to the reader

5.1 Structure of Assessments

Assessment in the first year is usually a mixture of coursework and written or online exams. The exact weighting varies between modules, so it's always worth checking the information provided on Blackboard or the module page at the start of the semester.

All first year modules also have a written in-person exam at the end of the semester. These typically take place during the university's official exam periods which is in January for Semester 1 modules and in the summer for Semester 2 modules.

The weighting of coursework and online/written exams can differ between modules and sometimes changes slightly from year to year, which is why it's important to check the module information once the course begins. Lecturers will normally explain the assessment structure in the first few weeks of the module as well.

5.2 Revision

5.2.1 Dealing with Stress

There is a particular kind of stress that arrives a few days before exams begin, and it's not really about content. It's about doubt.

You start asking questions that don't have neat answers. Have I revised enough? What if I've focused on the wrong topics? What if the paper goes badly? What if I've misunderstood something fundamental? What if… what if… what if…

The mind becomes very creative under pressure. At some point it stops being about mathematics and starts being about catastrophe.

Exams are famously rubbish at capturing understanding. And it's not just me. Exam stress, dread, and self doubt are incredibly common, especially among people who care deeply about their subject. But even knowing that, it's easy to feel alone when you're in the middle of it.

I remember feeling this intensely before my first January exams in Semester 1. I was convinced I hadn't done enough, that I was about to fail half my modules, and those thoughts seemed to be louder than any rational argument I could make.

I fondly remember speaking to my tutor about how stressed I was, and one of the first things he said to me was:

> *"It's gonna to be alright. I'm not saying you're gonna feel alright about it. I'm just saying it's gonna to be alright."*

I asked him how he knew, to which would response was, *"How? I don't know. It's a mystery."*

Then he showed me a **clip from Shakespeare in Love**, where Philip Henslowe says exactly that line when everything seems to be falling apart.

Watching an eighteen second YouTube video wasn't exactly the kind of reassurance I expected going into that conversation, but there was something oddly comforting about that phrase. Something about it stayed with me, and it wasn't because it removed the stress (it didn't) but rather that it removed the demand to know, to predict, to control. It wasn't saying I wouldn't worry, but that uncertainty doesn't mean catastrophe.

Later that evening, I printed out the script from Shakespeare in Love and put it on my wall. Since then, every time my mind started racing or I felt like I was catastrophising scenarios, I would look at it and smile, slightly against my will.

I don't think the lesson here is that exams should feel easy, or that stress can be eliminated entirely. The lesson, at least for me, was simpler than that.

You revise as well as you can, sit the paper, and attempt the questions in front of you with whatever you can bring to mind that day.

Going in with that mindset feels very different from walking into the exam convinced you're either going to ace everything, or on the flip side convinced you're going to fail everything. Both extremes are unhelpful: one sets you up for pressure, the other for panic.

Whereas if you accept that you don't know exactly how it will go, and that you might surprise yourself in either direction, you can focus on thinking clearly instead of predicting the outcome.

And once the paper leaves your hands, some part of it really is a mystery.

5.2.2 Comparison

Comparison is one of the fastest ways to drain your confidence during revision.

You will overhear conversations about how much or how many hours someone else has done, see people in the library morning to night, or someone say how they've completed every past paper twice.

None of that gives you actual useful information. You don't know how well they understood the material, or what their starting point was, or even whether what they are telling you is the entire truth or only a fraction of it.

The only revision that matters is yours. There will always be someone who appears more prepared, more calmer, or understands things instantly. Just because someone else appears better or smarter than you, doesn't make you any more or less good. Someone else understanding a concept quickly does not make you any less capable.

The person who seems completely calm might also be unsure, and the student who answers questions confidently in lectures might have struggled with a different topic entirely. Nobody every has it 'all worked out'. Comparison is always based on partial information, you rarely get to see the full picture. And during revision when anxiety and adrenaline are already elevated, comparison amplifies doubt without adding insight.

Mathematics is not a race in a straight line. Intellectual growth is personal, and often invisible. So let other people exist, let them be brilliant and revise in their

way. You task is not to outperform everyone, but to engage the best you can with the material in front of you.

5.3 Results

During exams you're busy thinking, writing, calculating, trying to make sense of questions under time pressure. But once the results come out, the thinking changes. You're no longer solving problems but instead interpreting a number and trying to decide what it means about you.

The reaction to good results isn't always excitement, sometimes it's thoughts like 'Did I actually deserve that?' Or 'I didn't work hard enough to deserve that grade'. It's that strange feeling of somehow getting away with something.

If you've ever had that thought, you're not alone. Mathematics has a way of making people doubt themselves even when things go well. The subject constantly shows you how much more there is to learn, so it's easy to feel like you're always a step behind. But you're allowed to accept the result you earned, you're allowed to have done better than you expected.

The opposite reaction can happen too.

Sometimes you work incredibly hard and the result feels disappointing and that can be much harder to process. When you've put hours into something, it's easy to let one number turn into a judgement about your ability.

That voice can be very convincing in the moment. Remember that results are merely a snapshot of how things went in a particular exam, on a particular day, under particular conditions.

It's easy to believe that the world is like a neat function, that the effort your input always results in the outcome you want, but the world is uncertain and full of variables beyond our control. That doesn't mean your effort was wasted, it just means your effort isn't the only deciding factor, and it definitely doesn't mean your worth is defined by one exam or one outcome.

In first year especially, it's worth keeping a bit of perspective. The marks from first year don't count towards your final degree as long as you pass the modules. Now, that doesn't mean the year doesn't matter. It absolutely does, in fact, it's where a lot of the real learning happens. But it does mean the stakes aren't quite as catastrophic as your brain might try to convince you.

If things genuinely don't go according to plan, there are formal processes in place

like summer resits, special considerations, etc. It's always worth checking in with your tutor following results if you find yourself in that situation.

But the bigger thing to remember is that mathematics is a subject where you are constantly aware of what you don't know yet. That can make you feel like you're the only one still figuring things out, even when that's not true. Belonging in this subject isn't about being the fastest person in the room, but about continuing to think even when things don't click immediately.

TIME MANAGEMENT

> The bad news is time flies. The good news is you're the pilot
>
> *Michael Altshuler*

6.1 Structuring Time

There is a funny shift that happens in the first few weeks of university. No-one tells you what to do between lectures.

You might have three lectures in the morning and then nothing scheduled until the following afternoon. That looks like a lot of free time, but in reality, it's hours of unstructured time. And unstructured time has a habit of vanishing.

In my experience, time management with a Maths degree is less about squeezing in as many hours as you possibly can and more about preventing last minute panic. The problems one engages with in mathematics often requires uninterrupted stretches of thought. Ten minutes here and then rarely work for deeper problems.

One hour of undistracted thinking is worth more than three hours of broken attention.

If you have a two hour gap between lectures, don't assume it's automatically productive time. Decide in advance what you're going to go. Sometimes you might be tired and use that to relax and read a novel or a hit the gym, or maybe you didn't quite understand the last 10 mins of the previous lecture and want to review the notes.

It doesn't matter what you do, as long as it's intentional.

6.2 Overworking vs Avoidance

It can be surprisingly difficult to tell the difference between overworking and avoidance.

If you are avoiding starting because the work feels uncomfortable, that's one thing. Usually once you begin, the momentum builds.

However, if you're staring at the page feeling exhausted and unable to process even simple steps, that's something else.

Avoidance is usually about resistance to beginning, whereas burnout is about depletion. And being honest about which one you are experiencing is important. There is no pride in pushing through exhaustion just to prove you can, and equally, there is no growth in endlessly procrastinating out of fear.

If you find yourself consistently overwhelmed, talk to someone. That could be your personal tutor, the Student Hub wellbeing team, or even your friends.

University has systems in place because they know how common it is for students to struggle with these things. A degree is a long process, and it rewards consistency far more than intensity.

6.3 Life Outside the Degree

It can be quite easy for your degree to consume everything. And sometimes that feels good, but remember that university is not just a degree factory.

There are lots of clubs and societies, sports, random Wednesday evenings that end up in unexpectedly good memories. Making time and space for those things is not a distraction from your degree, but part of the experience.

I have always found that I think more clearly after stepping away from a problem having spent a considerable amount of time on it. Some of the best insights into problems and questions arrived while I was engaged in something else entirely like cooking or rowing.

There will be weeks where the workload increases and the social things need to take a slight step back, and there will also be weeks where you might say yes to more events outside of academics. Whatever it might be, balance is not something

you're expected to achieve once and then maintain perfectly, it requires constant adjustment.

If you feel constantly exhausted, something needs shifts. And on the flip side if you feel always disengaged from your work, something also needs changing.

Doing a maths degree can feel intense at times as it does demand genuine critical thinking. But it still is a degree, not your entire identity. So do the work the best you can, engage in it with the best of you ability and remember to close the notebook sometimes and have fun.

FUTURE

7.1 Careers

USEFUL LINKS

School of Mathematical Sciences

Student Hub

My Career

Blackboard

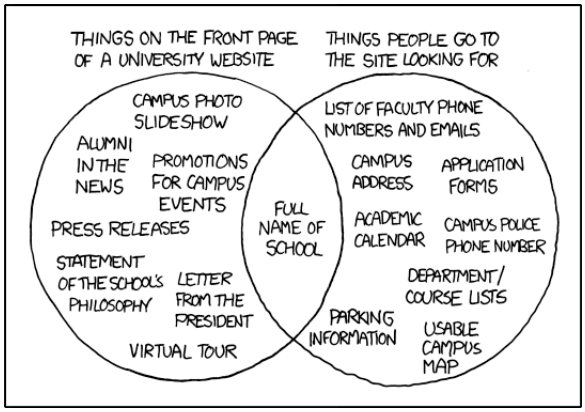

Figure 8.5: **University Website**

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